

# Machine Learning for Wireless Communication

## Assignment-2

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### Question 1: Part (b): SVM Classification at Different SNR Values

Six SVM classifiers were trained and tested separately at each SNR value using an 80/20 train-test split. An RBF kernel with  $C = 1$  was used for all classifiers. SVM-1 to SVM-5 use one feature at a time, while SVM-6 uses all five features together.

#### Classification Accuracy vs. SNR

The classification accuracy obtained for all six SVM classifiers is shown in Figure 1.

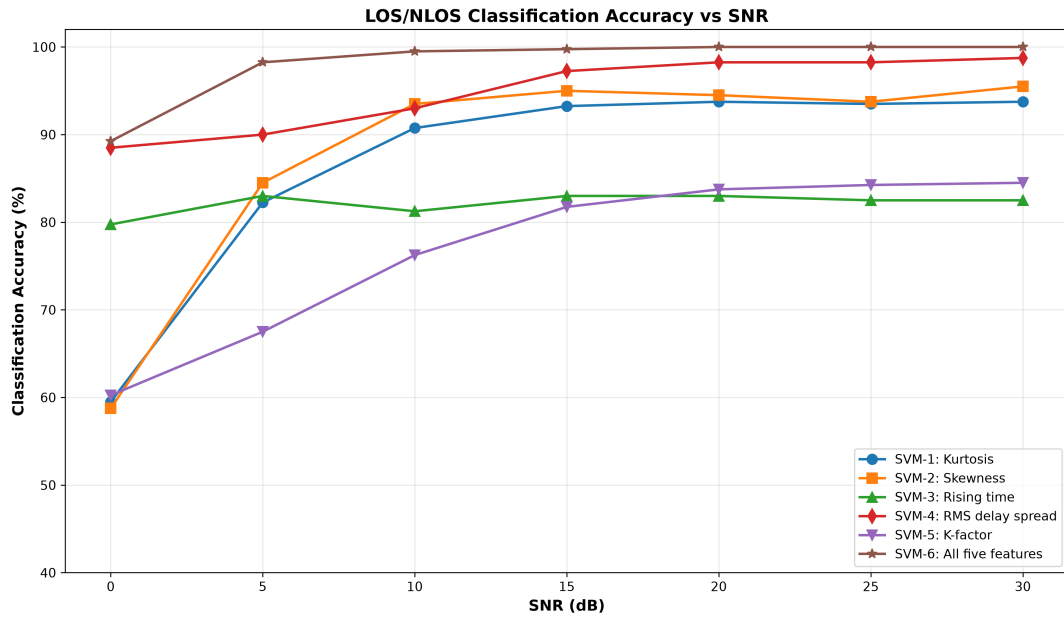


Figure 1: Classification accuracy of the six SVM classifiers as a function of SNR.

#### 1. Best Performing Single Feature

Among the five individual features, the RMS delay spread (SVM-4) gives the best overall classification performance. Its accuracy increases with SNR and reaches approximately 98% at high SNR values.

This can be explained physically because LOS channels generally contain a strong direct path with relatively small multipath dispersion, whereas NLOS channels contain more scattered multipath components and therefore tend to have a larger delay spread. Hence, RMS delay spread provides good separation between LOS and NLOS channels.

## **2. Effect of Combining the Features**

Combining all five features in SVM-6 generally improves the classification accuracy compared with using a single feature alone. The improvement is particularly noticeable at low-to-medium SNR, where individual features are more affected by noise.

The largest improvement occurs around 5 dB SNR. This is because the five features capture different properties of the channel, such as amplitude distribution, temporal structure, multipath dispersion, and dominant-path strength. Combining these complementary properties allows the SVM to form a more effective decision boundary.

At high SNR, the improvement becomes smaller because the individual features themselves become more reliable and already provide strong separation between LOS and NLOS channels.

## **3. Feature Performance at Low SNR**

Kurtosis is likely to perform poorly at very low SNR, even though it has physical significance for distinguishing the channel characteristics. Higher-order statistical features are sensitive to noise, and at low SNR the noise can significantly distort the amplitude distribution used to calculate kurtosis.

Therefore, the reliability of kurtosis decreases when the noise level is high. In comparison, RMS delay spread remains a relatively strong feature at low SNR in this experiment.

## **Part (c): Fixed-SNR Training and Generalisation**

In Part (c), SVM-6 was trained using data at only 25 dB SNR. The resulting fixed model was then tested at all SNR values from 0 to 30 dB. Its performance was compared with a model that was retrained at each test SNR.

### **Training at 25 dB vs. Retraining at Each SNR**

The comparison between the two training strategies is shown in Figure 5.

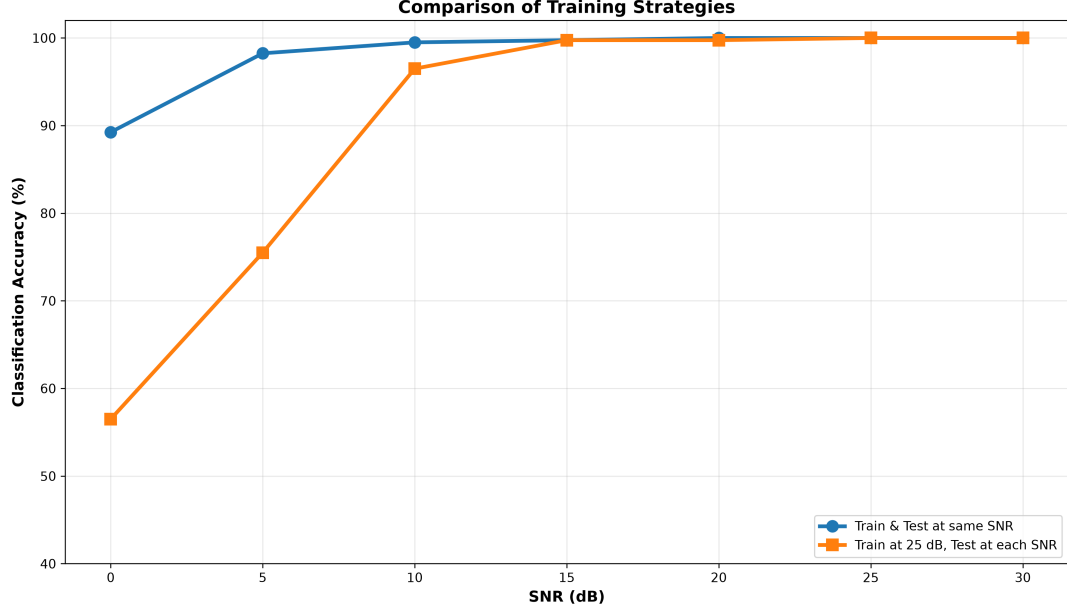


Figure 2: Comparison of a fixed SVM-6 trained at 25 dB with an SVM-6 retrained at each test SNR.

### 1. Agreement at High SNR

At high SNR values, the two curves become very close to each other. This shows that a model trained at 25 dB can generalise well to other high-SNR conditions.

The reason is that at high SNR the effect of noise on the extracted features is small. Therefore, the feature distributions become more stable across different high-SNR conditions, allowing the fixed model to maintain high classification accuracy.

### 2. Performance at Low SNR

At low SNR, the model retrained at the same SNR performs significantly better than the model trained only at 25 dB. For example, at 0 dB, the matched-SNR model achieves approximately 89% accuracy, whereas the fixed 25 dB model achieves approximately 59% accuracy.

The fixed high-SNR model does not generalise well to low SNR because the features observed at low SNR are strongly affected by noise. The model trained at 25 dB mainly learns the feature distributions corresponding to relatively clean channel observations. At low SNR, these distributions are distorted by the increased noise, resulting in reduced classification accuracy.

Thus, for low-SNR operation, training data that represents the actual operating SNR can provide significantly better performance.

### 3. Training at Low SNR and Testing at All SNRs

Training at a low SNR such as 0 dB and then testing across the complete SNR range would likely produce a different performance curve.

A model trained at 0 dB would learn feature patterns that are strongly influenced by noise. When the same model is applied at high SNR, the noise characteristics present during training would largely disappear, causing a mismatch between the training and testing feature distributions.

Therefore, its high-SNR performance could be lower than that of a model trained at a higher SNR. This suggests that the training SNR should be representative of the intended operating conditions. Alternatively, training with data from multiple SNR values could provide better generalisation across a wide range of SNR conditions.

## Question 2: Constellation Demodulation & Clustering using K-means

### Part (b): K-means Clustering

The K-means algorithm was applied to the received 16-QAM symbols at 25 dB SNR. The Cartesian coordinates  $(R_x, R_y)$  were used for the main K-means evaluation. The number of clusters was varied from  $K = 2$  to  $K = 20$ , with `n_init=10` and `random_state=42`.

#### 1. Inertia and Silhouette Analysis

Figure 3 shows the within-cluster sum of squares (inertia) and the silhouette coefficient for different values of  $K$ .

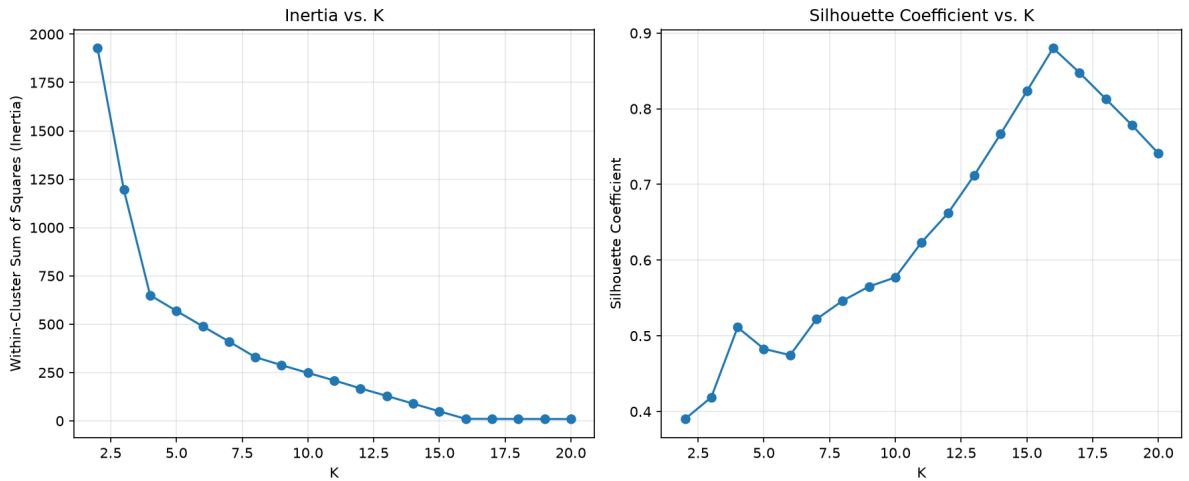


Figure 3: Inertia and silhouette coefficient versus the number of K-means clusters at SNR = 25 dB.

The inertia decreases rapidly as  $K$  increases. A major reduction is observed up to  $K = 16$ , after which the reduction becomes very small. The silhouette coefficient also reaches its maximum value of approximately 0.880 at  $K = 16$ . Therefore, both measures support choosing

$$K = 16.$$

This is also consistent with the fact that the 16-QAM constellation contains 16 distinct symbol points.

#### 2. $K = 16$ Constellation Clustering

Using  $K = 16$ , K-means was applied to the Cartesian coordinates of the received symbols at 25 dB SNR. The resulting clusters and their centroids are shown in Figure 4.

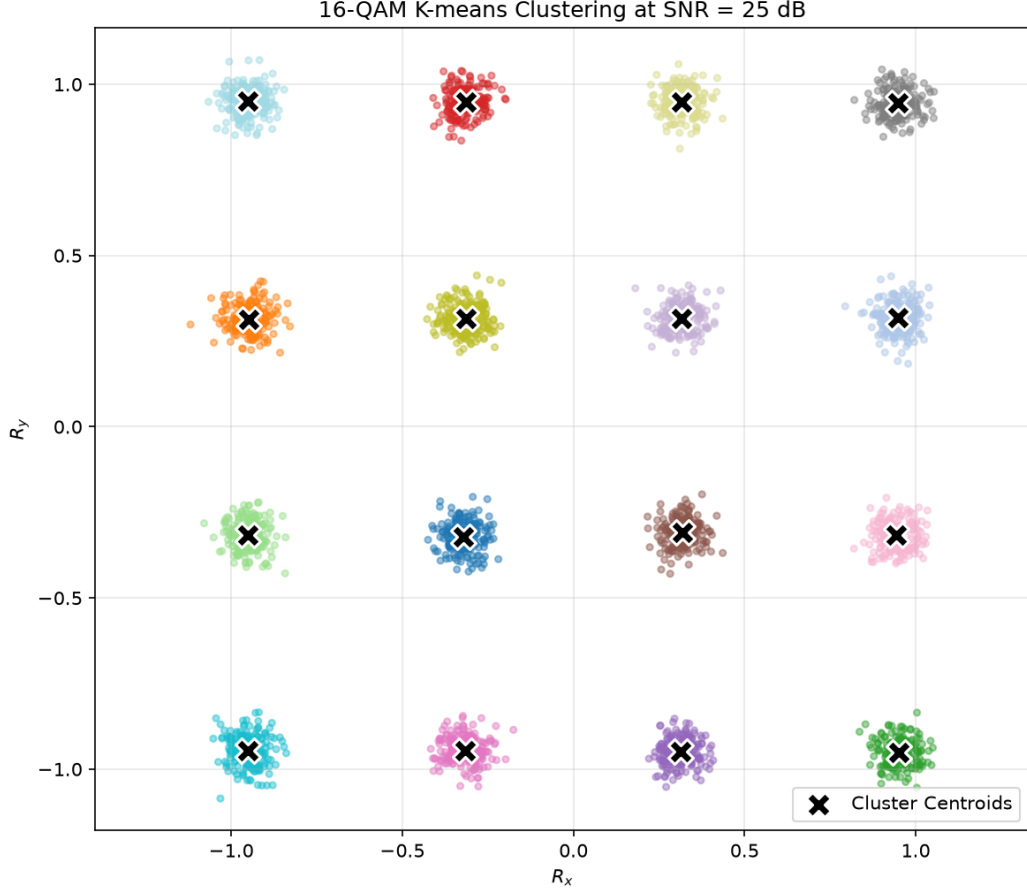


Figure 4: K-means clustering of the received 16-QAM constellation at SNR = 25 dB. The black crosses represent the cluster centroids.

The received points form 16 compact groups around the expected 16-QAM constellation locations. The centroids are located close to the centres of these groups, indicating that K-means successfully identifies the underlying constellation structure.

### 3. Cluster Purity for Different Feature Sets

The cluster purity obtained for the three feature representations is shown in Table 1.

Table 1: Cluster purity at SNR = 25 dB for different feature sets.

| Feature Set             | Cluster Purity (%) |
|-------------------------|--------------------|
| $(R_x, R_y)$            | 100.00             |
| $(r, \theta)$           | 52.22              |
| $(R_x, R_y, r, \theta)$ | 100.00             |

The Cartesian representation  $(R_x, R_y)$  gives the best performance, with a cluster purity of 100%. The combined representation also achieves 100% purity. In contrast, the polar representation  $(r, \theta)$  gives a much lower purity of 52.22%.

The Cartesian representation is well suited to 16-QAM because the constellation is defined as a two-dimensional rectangular grid in the in-phase and quadrature components. Euclidean

distance in the  $(R_x, R_y)$  plane therefore directly represents the geometric separation between constellation points.

The polar representation can distort this geometry because equal changes in phase do not correspond to equal Cartesian distances at different amplitudes. In addition, phase is a periodic quantity, so points near the phase boundary can appear artificially far apart when ordinary Euclidean distance is applied directly to  $(r, \theta)$ .

#### 4. Why Cartesian Coordinates Work Well for QAM

16-QAM symbols are arranged on a rectangular grid in the in-phase and quadrature dimensions. Thus, the Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(R_{x,1} - R_{x,2})^2 + (R_{y,1} - R_{y,2})^2}$$

preserves the natural geometry of the constellation. Points belonging to the same transmitted symbol form compact groups, while different symbols are separated by approximately equal grid spacings.

Hence, K-means using Cartesian coordinates can identify the 16 constellation clusters effectively.

### Part (c): Adaptive K-means vs. Fixed Template

For Part (c),  $K = 16$  was used. Two strategies were compared:

**Strategy 1:** K-means is retrained separately at every SNR.

**Strategy 2:** K-means is trained only at 25 dB and the resulting centroids are kept fixed for all test SNR values.

The resulting cluster purity curves are shown in Figure 5.

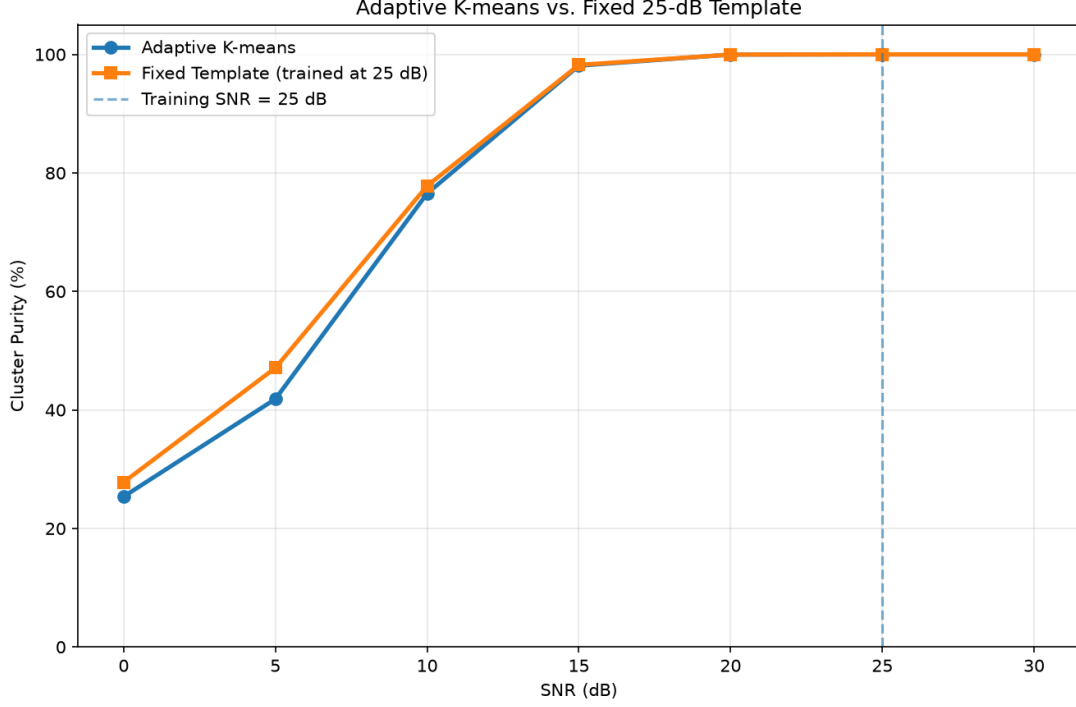


Figure 5: Cluster purity versus SNR for adaptive K-means and a fixed template trained at 25 dB.

### 1. High-SNR Behaviour

At high SNR, the two curves become almost identical. The adaptive and fixed-template approaches both reach approximately 100% cluster purity from 20 dB onwards.

This occurs because the effect of AWGN becomes small at high SNR. Consequently, the received points remain close to the ideal 16-QAM constellation, so the centroids learned at 25 dB remain suitable at other high SNR values.

### 2. Low-SNR Behaviour

At low SNR, both methods show a significant reduction in cluster purity. The fixed template performs slightly better than the adaptive K-means model for the generated dataset. At 0 dB, the adaptive model achieves 25.38% purity, while the fixed template gives 27.81%. At 5 dB, the corresponding purities are 41.84% and 47.09%.

The low purity is caused by the large amount of AWGN, which spreads the received points and causes overlap between neighbouring constellation clusters. The fixed template can perform slightly better in this experiment because its centroids remain anchored near the known 16-QAM constellation locations, whereas independently retrained K-means centroids can shift in response to the noisy samples.

### 3. Training at Low SNR and Testing at High SNR

If K-means were trained at a very low SNR and then used at high SNR, the result would be different. The centroids learned from the noisy low-SNR data would be displaced and would reflect the distorted constellation.

When the same centroids are applied to high-SNR data, the received points are much closer to the ideal constellation points, producing a mismatch between the learned template and the test distribution. Therefore, the fixed low-SNR template could give lower purity than a template trained at a higher SNR.

Overall, the results show that a fixed template trained at 25 dB generalises very well to high-SNR conditions, while performance at low SNR is strongly affected by noise.